

A Paired Evaluation of Clean Accuracy and Corruption Sensitivity in Quantized Object Detection

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ABSTRACT

Robustness comparisons between quantized object detectors can conflate two effects: accuracy differences that already exist on clean inputs and changes associated with image corruption. We study this problem with a paired four-arm evaluation in which INT8 and FP8 TensorRT treatments share condition-specific inputs and image-bootstrap draws, while clean accuracy, corrupted accuracy, and the change in the between-treatment gap are reported separately. Across six selection-disjoint VOC/KITTI holdout blocks, FP8 exceeds INT8 by 1.60 AP points on original-clean images and by 1.05 AP points under corruption (95% percentile interval [0.91, 1.16]). After removing the clean gap, however, the mean interaction is -0.55 AP $[-0.78, -0.29]$, showing that a detector can retain higher corrupted accuracy while its initial advantage contracts. In a broader four-dataset exploratory grid, raw and clean-adjusted point signs disagree in 56 of 144 conditions. Controlled clean-input substitution, relative-retention analysis, covariance audits, seed and corruption-realization studies, and a prediction-level reconstruction quantify how control choice, scale, and pairing affect interpretation. The study provides an auditable evaluation protocol for fixed quantized detector treatments and shows why absolute corrupted performance and clean-adjusted sensitivity should be reported together.

1. Introduction

Object detectors are increasingly deployed under two simultaneous constraints: reduced numerical precision and imperfect visual inputs. Post-training quantization (PTQ) lowers inference cost without repeating end-to-end training, while blur, noise, fog, and compression alter the evidence available to the detector. Evaluating both effects together is therefore a system-level measurement problem rather than a comparison of datatype names alone [1, 2].

A central ambiguity is that the raw performance gap under corruption contains any accuracy difference that was already present on clean images. Suppose FP8 starts 1.6 AP points above INT8 and remains 1.0 AP point above it after corruption. The corrupted-input ranking still favors FP8, but the advantage has contracted by roughly 0.6 AP. These statements answer different questions: one concerns remaining task performance, and the other concerns corruption-associated change relative to the clean baseline. The distinction matters most when clean quantization errors are unequal or when both detectors approach very low AP, where a shrinking gap can coexist with unusable recognition. Existing corruption benchmarks already motivate reporting absolute and clean-normalized performance [3]; the open practical question is how to make the corresponding

comparison between two executable quantized treatments paired, explicit, and auditable.

We address this question for recorded ModelOpt/TensorRT INT8 and FP8 object-detection treatments. Each treatment includes quantizer placement, calibration or scale construction, graph transformations, mixed-precision behavior, engine building, preprocessing, decoding, and runtime settings. We therefore do not interpret INT8 and FP8 as isolated causal factors. Instead, we compare the fixed executable treatments under a four-cell design: INT8/FP8 crossed with clean/corrupted inputs. The clean and corrupted treatment gaps are computed on shared images, and their difference is evaluated inside common image-bootstrap draws. Absolute corrupted AP and clean-to-corrupted loss are reported before the interaction, with relative retention used as a scale sensitivity rather than as a substitute estimand.

The empirical study contains two complementary scopes. A 144-cell grid spans COCO, VOC, KITTI, and TT100K; three YOLO11 capacities; four corruptions; and three severity levels. Because historical VOC/KITTI evaluation images participated in checkpoint selection, this grid is treated as exploratory. The strongest evidence comes from six retrained VOC/KITTI blocks evaluated on final images excluded from checkpoint selection. Additional analyses examine clean-control substitution,

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covariance preserved by pairing, corruption realizations, training/calibration seeds, class support, runtime, and portability of the quantization recipe to RT-DETR and RetinaNet.

The main contributions are:

- **A paired measurement protocol for quantized detector robustness.** It separates clean treatment fidelity, remaining corrupted accuracy, and corruption-associated change in the FP8–INT8 gap, with explicit invariants for inputs, execution artifacts, decoding, and common-image uncertainty.
- **Selection-disjoint evidence that the absolute and clean-adjusted conclusions can differ.** Across six final VOC/KITTI blocks, FP8 retains a positive mean corrupted advantage of 1.05 AP while the clean-adjusted interaction is -0.55 AP; a prediction-level example reconstructs the four APs and paired interval from retained predictions.
- **An empirical audit of the choices that change interpretation.** The broader grid, control-substitution experiment, relative-retention sensitivity, covariance audit, seed/realization analyses, and recipe-portability stress tests show which conclusions depend on the estimand, control definition, pairing, and treatment quality.

The primary empirical claim is deliberately scoped to the YOLO11 TensorRT treatments for which INT8 and FP8 are both meaningful executable comparisons. Cross-family RT-DETR and RetinaNet experiments are retained as stress tests because the transferred INT8 recipe is not clean-competitive; they are not used to infer a universal ranking of numerical formats.

2. Related work

2.1. Post-training quantization and executable 8-bit detection

PTQ estimates or reconstructs numerical ranges from representative data without full retraining. Foundational methods include range calibration, data-free surrogates, adaptive rounding, and local reconstruction [4–10]. Detector-specific PTQ methods further adapt calibration or reconstruction objectives to classification and localization outputs, including PD-Quant, DetPTQ, Reg-PTQ, and InlierQ [11–14]. These methods mainly optimize clean-input fidelity or deployment cost; our concern is how already constructed quantized detectors should be compared after input degradation.

INT8 has a mature integer-inference path, while FP8 provides a wider dynamic range through explicit exponent bits and has become practical on recent accelerators [15–18]. FP8 PTQ and flexible-format studies show that the preferred 8-bit representation depends on architecture, layer statistics, and hardware

support [19, 20]. Direct FP8–INT8 comparisons likewise show that accuracy and hardware efficiency need not favor the same format [21]. For executable detector engines, the nominal format is therefore incomplete: Q/DQ placement, calibration, graph rewriting, fallbacks, builder settings, preprocessing, and decoding all contribute to the observed treatment. We record and verify these components instead of attributing the full difference to datatype semantics.

2.2. Corruption robustness and quantized models

Synthetic corruption benchmarks provide controlled tests of noise, blur, visibility changes, and image-processing artifacts [22]. Robustness conclusions can depend on training augmentation and task structure [23]; dense-prediction studies report detector- and segmentation-specific failure modes under weather, sensor noise, blur, and camera distortion [3, 24, 25]. In object detection, Michaelis et al. report mean performance under corruption and relative performance normalized by the clean baseline [3]. These measures motivate the central distinction in this work: remaining corrupted accuracy and change relative to clean accuracy are related but non-equivalent targets.

Quantized-model robustness has been studied through clean-to-degraded performance drops, adaptive quantization, and out-of-distribution correction. Karimov et al. evaluate quantized YOLO detectors using absolute and relative degradation measures across precision and calibration settings [26]. TTAQ adapts PTQ under continuous domain change, while Recti-Q targets out-of-distribution quantized perception through feature correction [27, 28]. Adversarial studies similarly show that apparent robustness conclusions can change with the quantization pipeline [29]. Our setting differs in that the deployed treatments are fixed: no test-time adaptation is performed, and the measurement target is the paired change in the between-treatment gap.

2.3. Positioning of the present study

Clean adjustment itself is established. Relative mCE subtracts clean error, rPC normalizes corrupted performance by clean performance, and effective robustness models the relationship between clean and shifted accuracy across models [3, 22, 30]. We use a simpler fixed-treatment interaction: the corrupted FP8–INT8 AP gap minus the clean FP8–INT8 AP gap. The methodological contribution is not the subtraction algebra, but the operationalization of this contrast for executable quantized detectors: shared inputs within condition, common image resampling across all four arms, explicit clean-control semantics, absolute-accuracy guardrails, and artifact-level verification. Table 1 summarizes the closest measurement settings.

Table 1

Closest comparison settings and the measurement target of this study. “Not established” means not documented in the inspected report, not proven absent.

Study	Main comparison	Clean/corrupted measurement structure	Relation to this work
Michaelis et al. [3]	Absolute and clean-normalized detector performance under common corruptions	Clean baseline plus corruption benchmark; quantized-engine pairing not studied	Establishes detector corruption metrics and clean normalization
Karimov et al. [26]	Quantized YOLO accuracy and degradation across precision/calibration settings	Absolute and relative clean-to-degraded drops; no FP8 treatment	Closest quantized-detection robustness comparison
van Baalen et al. [21]	INT8/FP8 accuracy and hardware efficiency	Clean multi-task evaluation, including YOLOv5; corruption outside the reported comparison	Motivates direct FP8–INT8 executable comparison
This study	Change in a fixed FP8–INT8 AP gap under corruption	Four paired arms, explicit clean controls, shared-image uncertainty, absolute AP, and retention sensitivity	Auditable separation of clean fidelity from corruption-associated gap change

3. Methodology

3.1. Study design and treatment definition

Figure 1 summarizes the paired design. COCO uses the official Ultralytics YOLO11n/m/x checkpoints; VOC, KITTI, and TT100K use recorded `best.pt` checkpoints trained within the same YOLO11 family. Each checkpoint is exported to TensorRT engines. FP16 is used only as a consistency check. The main executable treatments are a high-precision TensorRT reference, INT8 with entropy calibration, and FP8. INT8 and FP8 denote complete recorded ModelOpt/TensorRT pipelines rather than isolated datatype effects: quantizer placement, Q/DQ coverage, calibration or scale construction, graph changes, builder settings, fallbacks, preprocessing, runtime, and decoding are part of the treatment.

Within a paired condition, clean or corrupted images are materialized before the precision branch. INT8 and FP8 therefore receive the same ordered image identifiers and encoded bytes for that condition. The study evaluates the recorded treatments; it does not estimate the best attainable implementation of either format.

3.2. Paired estimands and reporting order

Let d , m , b , and c index dataset, model, size endpoint, and corruption condition. The exploratory matched-clean control, denoted J95, is the source image after deterministic JPEG quality-95 encoding with chroma subsampling disabled and no named corruption. The final holdout replaces J95 by original-source clean. AP is evaluated on the $[0, 1]$ scale and reported here in AP points after multiplication by 100.

For clean or corrupted input x , define the direct FP8–INT8 gap

$$G_{d,m,b,x} = \text{AP}_{d,m,\text{FP8},b,x} - \text{AP}_{d,m,\text{INT8},b,x}. \quad (1)$$

The primary interaction is

$$\Delta E_{d,m,b,c} = G_{d,m,b,c} - G_{d,m,b,\text{J95}}. \quad (2)$$

For the final holdout, original-source clean replaces J95 in Eq. (2). Positive ΔE means that the signed FP8–INT8 gap is larger after corruption; negative ΔE means that it is smaller. The contrast is descriptive and conditional on the fixed treatments; it is not a causal difference-in-differences estimator. The FP32 reference cancels algebraically from this direct contrast and is retained only for treatment-verification and reference-relative diagnostics.

Absolute performance is reported separately. For treatment p ,

$$D_{d,m,p,b,c} = \text{AP}_{d,m,p,b,\text{J95}} - \text{AP}_{d,m,p,b,c}, \quad (3)$$

so $\Delta E = D_{\text{INT8}} - D_{\text{FP8}}$. A negative interaction can therefore arise when FP8 loses more of an initial clean advantage while still retaining higher corrupted AP. It can also occur when both treatments fail: if both corrupted AP values approach zero while the clean gap is fixed,

$$\lim_{\text{AP}_{\text{INT8},c}, \text{AP}_{\text{FP8},c} \rightarrow 0} \Delta E = -G_{\text{J95}}. \quad (4)$$

For this reason, corrupted AP and signed loss D are an *absolute-accuracy guardrail*, not a statistical floor test or deployment threshold.

As a secondary scale sensitivity, each treatment’s relative retention is

$$R_{d,m,q,b,c} = 100 \frac{\text{AP}_{d,m,q,b,c}}{\text{AP}_{d,m,q,b,\text{J95}}}, \quad \Delta R = R_{\text{FP8}} - R_{\text{INT8}}. \quad (5)$$

Paired Evaluation of Quantized Detector Robustness

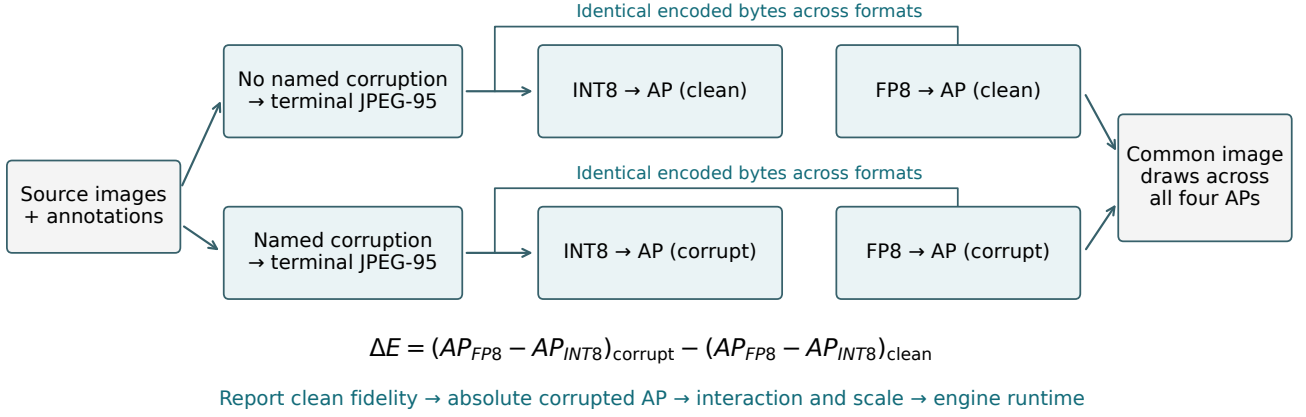


Figure 1: Paired four-arm evaluation used in the exploratory grid. Clean and corrupted inputs are materialized before the precision branch so that INT8 and FP8 receive identical encoded bytes within each condition. The primary interaction subtracts the clean FP8–INT8 gap from the corrupted gap inside common image-bootstrap draws. The final VOC/KITTI holdout uses original-source clean inputs; a controlled follow-up evaluates the original/JPEG-95 clean-control substitution with fixed engines and one runner.

Table 2

Measurement contract for a paired quantized-detector comparison.

Question	Quantity	What it isolates	Interpretation boundary
Clean treatment fidelity	G_{clean} and reference gaps	Difference before the named corruption	Does not measure corruption sensitivity
Remaining task performance	AP_c and D	Corrupted accuracy and clean-to-corrupted loss	Application usefulness needs an external tolerance
Change in treatment gap	$\Delta E = G_c - G_{\text{clean}}$	Corruption-associated change after removing the clean gap	Not a universal format ranking
Scale sensitivity	ΔR	Difference in fractions of clean AP retained	A different denominator-dependent estimand
Efficiency	Engine size, latency, throughput	Cost of the recorded executable treatment	Engine-only timing is not end-to-end latency

ΔR is reported in percentage points and changes the estimand by normalizing each treatment to its own clean AP. Size-specific interactions are secondary and are detailed in S1; COCO/VOC/KITTI area endpoints and TT100K height endpoints are never pooled.

Table 2 defines the interpretation order used throughout the paper. Clean fidelity is checked first, then remaining corrupted accuracy, then the clean-adjusted interaction and its scale sensitivity. Engine efficiency is interpreted only after these accuracy diagnostics.

3.3. Evidence layers, datasets, and holdout design

The exploratory grid contains 4 datasets $\times$ 3 YOLO11 capacities $\times$ 4 corruptions $\times$ 3 severities = 144 all-object interaction cells. Its balanced summary is the arithmetic mean over this finite grid, not an estimate weighted for an unspecified deployment population. COCO uses val2017 [31]; the transfer datasets use PASCAL VOC [32], KITTI [33], and

TT100K [34]. VOC and KITTI annotations are converted to the COCO evaluator, so the reported AP values are COCO-style measurements rather than official VOC/KITTI challenge scores. Dataset counts are shown in Table 3; S1 gives size-endpoint conventions and conversion details.

The historical four-dataset grid is exploratory because VOC2007 test and the recorded KITTI validation partition participated in checkpoint selection. The final rerun freezes new partitions before retraining with seed 20260818. VOC uses 8,218 train2007+train2012 images for training, 2,510 val2007 images for selection, and 5,823 val2012 images for final evaluation. KITTI reserves 1,496 images for selection, then a deterministic label-aware 20% subset of the remaining images for final evaluation (1,197), leaving 4,788 for training. Calibration uses training data only. The final images are image-disjoint from checkpoint selection, although KITTI is not verified scene- or drive-disjoint. The resulting six dataset–model blocks

Table 3

Frozen evaluation datasets. Counts refer to the converted annotations consumed by the evaluator.

Dataset	Split	Images	Objects	Declared/observed classes	Input px	Size strata
COCO	val2017	5000	36781	80/80	640	area: $[0, 32^2]$ / $[32^2, 96^2]$ / $\geq 96^2$
VOC	VOC2007 test	4952	12032	20/20	640	area: $[0, 32^2]$ / $[32^2, 96^2]$ / $\geq 96^2$
KITTI	validation	1496	8128	8/8	640	area: $[0, 32^2]$ / $[32^2, 96^2]$ / $\geq 96^2$
TT100K	test	3067	8181	221/136	1280	height: XS/S/M/L/XL

and 72 corruption cells form the paper’s strongest post-selection evidence.

The final holdout uses original-source clean bytes. Because the exploratory grid uses the JPEG-95 matched-clean control, the two evidence layers estimate slightly different clean-control targets. A controlled clean-input follow-up therefore evaluates original-source and JPEG-95 clean images with the same retained engine and runner in the six holdout blocks, plus two separately reported TT100K diagnostics. This isolates the clean-control substitution within the recorded execution path; it does not establish codec equivalence.

3.4. Precision treatments, corruptions, and execution controls

Engines use TensorRT 11.1.0.106 [35] on an RTX 5090 with a 4096 MiB builder workspace. ModeLOpt provides the recorded INT8 and FP8 quantization pipelines [36]. The final high-precision reference explicitly disables TF32 and gates source/ONNX Runtime/FP32 TensorRT agreement at 0.1 AP point and FP32/FP16 TensorRT agreement at 1.0 AP point. These are consistency checks rather than numerical-equivalence proofs. Artifact verification is graded: retained hashes, Q/DQ graph evidence, build records, engine/prediction bindings, and unrecorded fields are reported separately in S1. Q/DQ counts describe graph structure, not runtime-kernel precision percentages.

The four corruption families use severity labels 1/3/5: Gaussian noise ($\sigma = 8/20/40$ on the 0–255 scale), motion blur (5/13/23-pixel kernels), fog (strength–radius pairs (0.18, 6)/(0.38, 12)/(0.62, 20)), and JPEG quality 70/40/15. Exploratory corrupted images and their matched-clean controls receive deterministic terminal JPEG-95 encoding with chroma subsampling disabled. Images are materialized once before inference with dimensions and ordered identifiers preserved. YOLO inference uses fixed letterbox geometry, confidence 10^{-3} , class-wise NMS IoU 0.7, a 300-detection decoder cap, and COCOeval maxDets = 100.

For the controlled clean-input experiment, original-source and JPEG-95 clean images are passed through the same engine, preprocessing, decoder, threshold, NMS, annotations, class map, and ordered image list. For each block, the control-induced change in the interaction is

$S = \Delta E_{J95} - \Delta E_{\text{original}} = G_{\text{original}} - G_{J95}$ while the corrupted arms are held fixed. One estimate is reported per block; reusing the same clean gap across multiple corruptions is not treated as repeated codec evidence.

3.5. AP evaluation and paired uncertainty

AP is evaluated over IoU thresholds 0.50–0.95. The image is the resampling unit [37]. Within each dataset and evidence layer, one deterministic schedule of 2,000 image-index bootstrap draws is reused across the related clean, corrupted, INT8, and FP8 arms. The custom AP accumulator indexes per-image COCOeval entries by the sampled position list without deduplicating repeats; it does not pass duplicate image IDs back to COCOeval.evaluate(). Stable score sorting preserves tie order, and the interaction is calculated inside each common draw before aggregation. The shared schedule preserves the observed covariance between the clean and corrupted treatment gaps.

Percentile intervals therefore quantify uncertainty from the finite evaluation images conditional on fixed checkpoints, calibration lists, engines, corruptions, decoding, and evaluation splits. They do not propagate training-seed, calibration-seed, engine-build, dataset-selection, or corruption-population uncertainty, and individual-cell intervals are not multiplicity-adjusted. The point-sign diagnostic compares the raw corrupted gap G_c with ΔE ; a sign disagreement is interpreted as a change of estimand, not as two contradictory statistical decisions. A retained-draw covariance audit compares the paired variance with the variance obtained after analytically removing only the observed clean–corrupted gap covariance. S1 also reports a fixed-class-universe Bayesian-bootstrap sensitivity for TT100K [38].

3.6. Sensitivity analyses and secondary diagnostics

Three targeted analyses probe conditionality beyond finite-image resampling. First, YOLO11m is evaluated on frozen, label-aware 512-image subsets of VOC, KITTI, and TT100K under three fixed corruption labels; stochastic fields and blur angles are nested across severity in this sensitivity, whereas deterministic JPEG repeats across labels. Second, a 3×3 crossing of training and calibration seeds is evaluated for YOLO11m on the same transfer datasets at severity 5.

Third, the 144-cell ledger is re-aggregated with image-count and object-count weights and after omitting each dataset or corruption family. These are bounded design sensitivities, not repeated estimates of one population parameter [39].

RT-DETR-L [40, 41] and RetinaNet [42, 43] are retained as quantization-recipe portability stress tests. Their transferred INT8 treatments lose substantial clean accuracy, so these blocks are not used as balanced cross-family FP8–INT8 comparisons. A build-policy-matched default-versus-aligned INT8 comparison in KITTI–RetinaNet, which reuses rather than rebuilds FP8, and TIDE AP50 diagnostics [44] are reported in S1 as secondary evidence, not mechanistic explanations of datatype behavior.

3.7. AI-assisted research workflow

OpenAI Codex was used as an author-supervised coding assistant for parts of experimental orchestration, analysis implementation, validation, and consistency checking. The hosted model snapshot was not retained in the historical experiment records. AI-generated code or suggestions were reviewed, tested, and edited by the authors; detector predictions and numerical measurements were produced by the recorded experimental software rather than by a generative-AI model. Use of generative AI for manuscript preparation is disclosed separately at the end of the article.

4. Results

4.1. Primary selection-disjoint holdout evidence

The delivered four-arm evaluator recomputes AP from annotations and predictions without relying on AP caches. Regression fixtures verify zero interaction for identical treatments or identical clean/corrupted predictions, sign reversal after treatment exchange, repeated-image multiplicity, and rejection of mismatched image or execution settings. As a prediction-level worked example, KITTI’s 1,197 final images with YOLO11m and fog severity 1 reproduce clean INT8/FP8 AP of 67.3281/70.5287 and corrupted AP of 63.5892/66.5094. FP8 therefore remains 2.9202 AP ahead under corruption, but the gap has contracted from 3.2007 AP on clean inputs, giving $\Delta E = -0.2805$ AP. The regenerated 2,000-draw interval is $[-1.2692, +0.7844]$.

Across all six final VOC/KITTI dataset–model blocks, FP8’s original-clean advantage averages 1.60 AP points and is positive in every block. Under corruption, the balanced FP8–INT8 gap remains positive at +1.05 AP with a paired 95% percentile interval of $[+0.91, +1.16]$. The clean-adjusted interaction, however, is -0.55 AP $[-0.78, -0.29]$. Dataset means are -0.43 AP for VOC and -0.67 AP for KITTI (Table 4). Thus the strongest holdout result

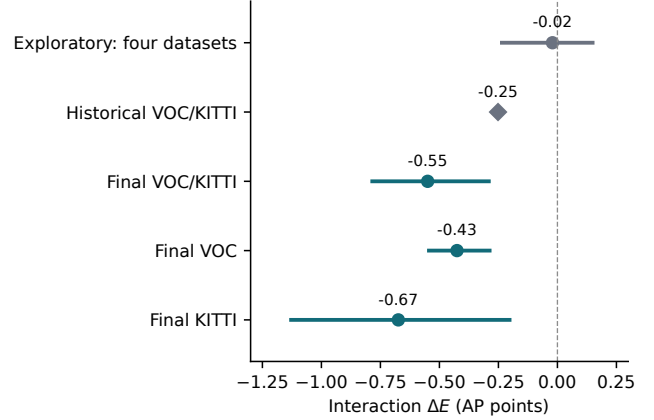


Figure 2: Selection-disjoint VOC/KITTI holdout evidence beside historical and exploratory context. Bars are paired-image percentile intervals within each stated scope; the historical two-dataset value is shown only as a contextual point estimate.

Table 4

Dataset summaries from the final holdouts using original-source clean controls. Effects and intervals are AP points from common paired image samples.

Dataset	Selection	Final	Cells	Mean ΔE	95% interval
VOC	2,510	5,823	36	-0.43	$[-0.54, -0.29]$
KITTI	1,496	1,197	36	-0.67	$[-1.13, -0.20]$

is not that FP8 becomes worse under corruption; it is that a positive absolute advantage becomes smaller after accounting for the advantage already present on clean images.

All six strict high-precision reference-parity blocks passed. The largest source-to-FP32-engine difference was 0.0404 AP and the largest FP32-to-FP16 difference was 0.2554 AP. Figure 2 keeps the final holdout separate from the historical and four-dataset exploratory summaries because their partitions, checkpoint histories, and clean-control definitions differ.

Table 5 reports the four AP components behind this interpretation. A post hoc point-sign diagnostic of the frozen holdout grid finds 52 positive raw corrupted gaps; 29 become negative after clean adjustment. Eight of those adjusted intervals lie entirely below zero and 21 cross zero. These counts describe how the estimand changes and are not multiplicity-adjusted discoveries.

4.2. Clean-control sensitivity and absolute corrupted accuracy

The controlled clean-input follow-up completed all 32 clean arms with fixed engines and execution settings. Across the six final-holdout blocks, replacing original-source clean by JPEG-95 clean changes the interaction by only +0.0574 AP on average, with paired interval $[-0.1839, +0.2518]$. Individual point shifts range from -0.2902 to $+0.7985$ AP and all six intervals cross zero (Table 6). The separately selected

Table 5

Original-source-clean final holdout: four AP values and contrasts (AP points). Corrupted AP is the equally weighted mean of twelve corruption–severity cells per block. $G_0 = A_{F,0} - A_{I,0}$, $G_c = \bar{A}_{F,c} - \bar{A}_{I,c}$, and $\Delta E = G_c - G_0$. Brackets give paired 95% percentile intervals from 2,000 draws; block and macro means precede percentiles. G_c intervals combine retained ΔE draws with original-clean AP draws recovered on the recorded historical schedule; historical NumPy/evaluator versions were not recorded. These are original-source estimates, not codec-matched estimates.

Dataset	Model	Original clean		Corrupted mean		G_0	G_c [95% CI]	ΔE [95% CI]
		INT8	FP8	INT8	FP8			
VOC	YOLO11n	53.41	53.84	37.13	37.69	+0.4274	+0.5610 [0.4368, 0.6855]	+0.1336 [-0.0757, 0.3982]
VOC	YOLO11m	56.26	57.91	40.34	41.19	+1.6546	+0.8498 [0.6888, 0.9554]	-0.8048 [-1.0208, -0.5717]
VOC	YOLO11x	58.43	58.66	42.64	42.27	+0.2291	-0.3762 [-0.4762, -0.2808]	-0.6053 [-0.7919, -0.4096]
KITTI	YOLO11n	58.33	60.53	41.12	42.44	+2.1946	+1.3196 [0.9552, 1.7035]	-0.8750 [-1.7850, 0.0994]
KITTI	YOLO11m	67.33	70.53	46.00	48.14	+3.2007	+2.1344 [1.7440, 2.4366]	-1.0662 [-1.9657, -0.1723]
KITTI	YOLO11x	68.90	70.80	46.74	48.55	+1.8938	+1.8128 [1.4292, 2.1585]	-0.0810 [-0.8485, 0.5881]
Primary equal-block macro (six blocks)							+1.0502 [0.9105, 1.1554]	-0.5498 [-0.7843, -0.2908]

Table 6

Controlled clean-only substitution (AP points). Original-source and JPEG-Q95 clean AP use the same retained engines and paired image draws. $S = (A_{F,0} - A_{I,0}) - (A_{F,95} - A_{I,95})$ is a change in clean gap, not an independently replicated corruption effect. Intervals are paired image-bootstrap percentile 95% intervals (2,000 draws), conditional on retained engines and encoded bytes.

Dataset	Model	Original clean		JPEG-Q95 clean		S	95% CI
		INT8	FP8	INT8	FP8		
Primary: six VOC/KITTI holdout blocks							
VOC	YOLO11n	53.41	53.84	53.32	53.75	+0.0011	[-0.2693, 0.2324]
VOC	YOLO11m	56.26	57.91	56.04	57.74	-0.0460	[-0.2434, 0.1774]
VOC	YOLO11x	58.43	58.66	58.26	58.59	-0.1011	[-0.3000, 0.0561]
KITTI	YOLO11n	58.33	60.53	58.40	60.61	-0.0182	[-0.8572, 0.9690]
KITTI	YOLO11m	67.33	70.53	67.80	70.20	+0.7985	[-0.0440, 1.5560]
KITTI	YOLO11x	68.90	70.80	68.61	70.79	-0.2902	[-0.9880, 0.4470]
Primary equal-block macro						+0.0574	[-0.1839, 0.2518]
Diagnostic: TT100K (not pooled)							
TT100K	YOLO11n	25.10	27.12	25.29	26.41	+0.8995	[-0.1497, 1.9384]
TT100K	YOLO11x	55.50	56.76	56.02	56.10	+1.1878	[-0.7891, 2.1028]

TT100K n/x diagnostics have larger point shifts (+0.8995 and +1.1878 AP) but wide intervals that also cross zero. The experiment therefore reduces concern that the six-block holdout result is driven by the clean codec choice, while providing no equivalence claim for codecs in general.

In the exploratory grid, FP8 has higher JPEG-95 clean AP than INT8 in all 12 dataset–YOLO combinations, by 1.40 AP on average. Under corruption, 282 of 288 treatment arms lose AP relative to their matched-clean input; the median signed loss is 5.51 AP and the mean is 15.24 AP. Thirty-five exploratory arms fall below the descriptive guide of 10 AP and

nine fall below 5 AP. These values are not deployment thresholds. They establish why a clean-adjusted interaction cannot substitute for reporting remaining task accuracy: a shrinking treatment gap can occur even when both detectors perform poorly. Dataset-level absolute summaries are shown in Table 7.

4.3. Exploratory landscape and decision impact

Across the 144 exploratory cells, the raw corrupted FP8–INT8 gap is positive in 134 cells and averages 1.38 AP. The 12 dataset–model clean gaps are also all positive and average 1.40 AP. After subtracting the matched-clean gap, mean ΔE is -0.02 AP and the point

Table 7

Absolute corrupted-accuracy guardrail from the exploratory YOLO ledger. The signed loss is $D = AP_{J95} - AP_c$. Rows are balanced descriptive summaries, not deployment thresholds or pooled leaderboard scores.

Dataset	Format	Clean AP	All-corrupt AP	Mean D	Severity-5 AP	Severity-5 D
COCO	INT8	46.010	34.911	11.099	25.185	20.826
COCO	FP8	47.110	35.824	11.286	25.667	21.443
VOC	INT8	65.732	49.246	16.486	35.478	30.255
VOC	FP8	66.828	50.300	16.529	36.023	30.805
KITTI	INT8	67.284	46.445	20.839	29.881	37.403
KITTI	FP8	69.797	48.498	21.299	31.504	38.293
TT100K	INT8	44.760	32.270	12.490	22.573	22.188
TT100K	FP8	45.636	33.752	11.884	23.467	22.169

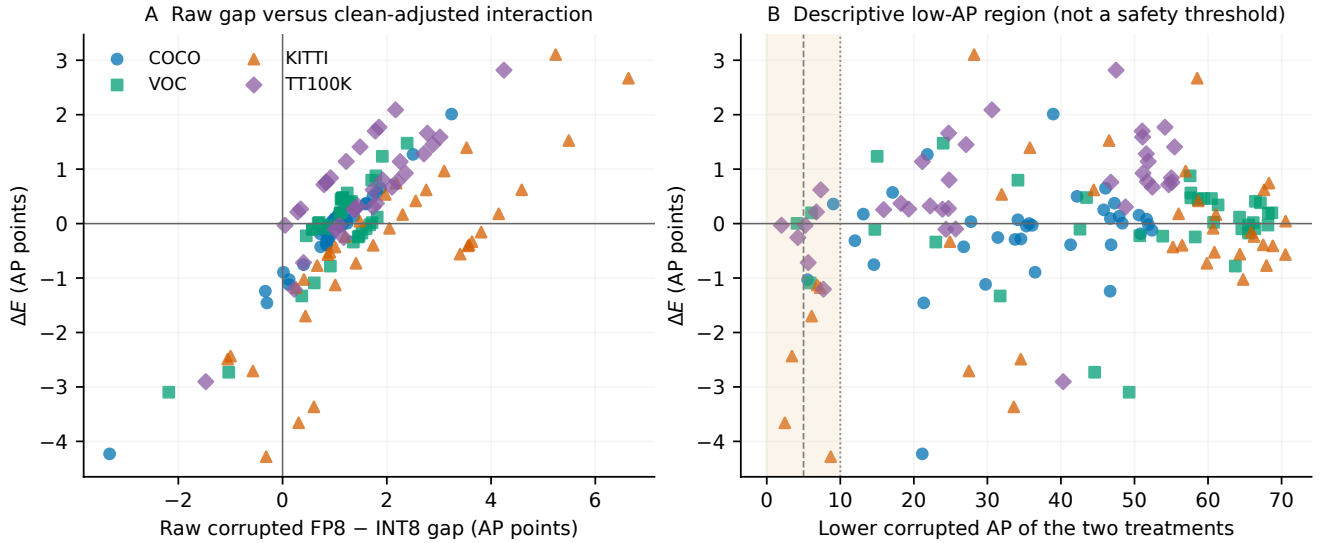


Figure 3: Why raw corrupted superiority and clean-adjusted sensitivity can differ. (A) The raw corrupted FP8-INT8 gap includes the pre-existing clean gap. (B) The same interaction values are shown against the lower corrupted AP of the two treatments; 5- and 10-AP lines are descriptive guides rather than deployment thresholds.

estimates split into 78 positive and 66 negative values. Fifty-six cells change from a positive raw gap to a negative clean-adjusted point estimate (Fig. 3A); 17 of their adjusted intervals lie entirely below zero and 39 cross zero. Twenty-two of the 56 disagreements have $|\Delta E| < 0.25$ AP and 36 have $|\Delta E| < 0.50$ AP, so the sign count should not be interpreted as 56 practically large reversals.

Figure 3B adds the lower corrupted AP of the two treatments. Eighteen cells have at least one treatment below 10 AP and five have at least one below 5 AP. This visualization makes the two required questions explicit: clean adjustment changes the comparison target, while absolute AP determines whether the remaining detector performance is meaningful for a particular application.

Table 8

All-object dataset summaries from the exploratory paired ledger.

Dataset	Cells	Mean ΔE
COCO	36	-0.19
VOC	36	-0.04
KITTI	36	-0.46
TT100K	36	+0.61

The balanced four-dataset mean $\overline{\Delta E}_{\text{grid}}$ is -0.02 AP with percentile interval $[-0.24, +0.15]$, but condition-level effects are heterogeneous: they range from -4.28 to $+3.10$ AP. Dataset means span -0.46 AP for KITTI to $+0.61$ AP for TT100K (Table 8). The near-zero macro therefore does not imply that individual conditions are insensitive to corruption or that a single format dominates across datasets.

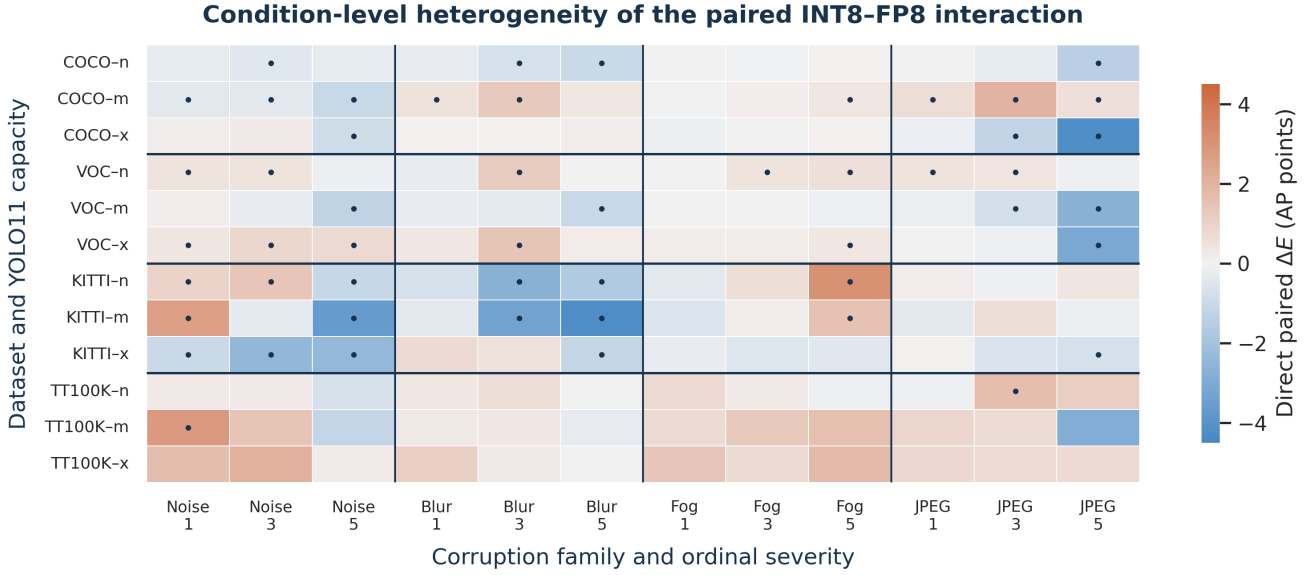


Figure 4: Exploratory interaction heterogeneity across dataset, YOLO11 capacity, corruption, and severity. Dots mark individual-cell percentile intervals that do not cross zero; these intervals are exploratory and are not adjusted for multiple comparisons.

Table 9

Empirical consequences of key protocol choices. The rows change either the scientific target, the uncertainty calculation, or the reporting context; they are not an algorithmic ablation leaderboard.

Component	What changes?	Observed consequence	Interpretation boundary
Subtract clean gap	Estimand: G_c to ΔE	56/144 exploratory point-sign changes	Not 56 multiplicity-adjusted discoveries
Preserve clean-corrupted covariance	Uncertainty, not point estimand	Zero-covariance SD larger in 127/144 cells; median ratio 1.124	Does not establish interval coverage
Report absolute corrupted AP	Adds task-performance context	282/288 exploratory arms lose AP; 35 below 10 AP	Descriptive cutoffs, not estimated failure thresholds
Use relative retention	Estimand and scale	18/144 exploratory and 7/72 holdout sign disagreements with ΔE	Denominator-dependent sensitivity
Clean-control substitution	Clean comparison target	Six-holdout mean shift +0.057 AP, interval crosses zero	No general codec-equivalence claim

4.4. Sensitivity to estimand, pairing, and design choices

Table 9 summarizes the main measurement consequences. Clean subtraction changes the estimand and produces the 56/144 exploratory point-sign disagreements above. Preserving clean-corrupted covariance does not change the point estimate, but setting that covariance to zero increases the estimated SD in 127 of 144 cells. The median zero-covariance/paired SD ratio is 1.124 (range 0.912–1.748), showing that shared-image sampling has a measurable and data-dependent effect on uncertainty.

The relative-retention contrast yields a different scale-dependent summary. Across the exploratory

grid, mean ΔE is -0.02 AP whereas mean ΔR is $+0.75$ percentage points, with 18 cell-level sign disagreements. Across the final holdouts, the corresponding means are -0.55 AP and -0.12 percentage points, with seven sign disagreements. KITTI’s final dataset mean changes from -0.67 AP on the additive scale to $+0.10$ percentage points on the retention scale. We keep ΔE primary because it directly answers how the between-treatment AP-point gap changes; ΔR answers how much of each treatment’s own clean AP is retained.

Other design sensitivities support conditional rather than universal interpretation. Equal-, image-count-, and object-count-weighted exploratory means

Table 10

Metric-scale sensitivity. ΔR is the FP8-minus-INT8 difference in the percentage of clean AP retained, using each evidence layer’s recorded clean control.

Scope	Cells	Mean ΔE (AP)	Mean ΔR (pp)	Sign differences
Exploratory grid	144	-0.02	+0.75	18
Exploratory: COCO	36	-0.19	+0.16	1
Exploratory: VOC	36	-0.04	+0.35	6
Exploratory: KITTI	36	-0.46	+0.49	6
Exploratory: TT100K	36	+0.61	+1.98	5
Final holdouts	72	-0.55	-0.12	7
Final: VOC	36	-0.43	-0.34	3
Final: KITTI	36	-0.67	+0.10	4

Table 11

TensorRT engine measurements on the RTX 5090. Values for each condition are medians of three runs; each row summarizes 12 corresponding YOLO conditions.

Precision	n	Size (MiB)	Latency (ms)	Throughput (qps)
FP32	12	79.90	2.434	410.6
INT8	12	24.07	0.789	1265.3
FP8	12	21.92	0.760	1315.7

are -0.02 , $+0.00$, and -0.09 AP; leave-one-dataset means range from -0.23 to $+0.13$ AP. Three corruption-realization labels produce mean interactions of -0.12 , -0.26 , and -0.32 AP in their targeted branch. In the 3×3 training/calibration seed study, the balanced mean is -1.11 AP, with training-seed marginals from -1.64 to -0.68 and calibration-seed marginals from -1.32 to -0.84 AP. The TT100K fixed-class-universe bootstrap preserves the negative height-macro direction but changes whether the overall interval crosses zero. Full conditionality tables and Monte Carlo checks are reported in S1.

4.5. Runtime and cross-family portability stress tests

Across the 12 corresponding YOLO conditions, the median of the condition-wise FP32/INT8 latency ratios is $2.96\times$, and the corresponding FP32/FP8 median is $3.13\times$. The median of the condition-wise INT8/FP8 ratios is $1.08\times$ with a range of 0.92 – $1.19\times$, so FP8 is not faster in every condition. These are engine-only measurements on one RTX 5090 and exclude image decoding, preprocessing, transfer, detector decoding, NMS, power, and energy (Table 11).

The RT-DETR-L and RetinaNet blocks are not balanced format comparisons. The transferred INT8 recipe has mean clean-reference gaps of 38.66 AP for RT-DETR-L and 22.53 AP for RetinaNet, with mean corrupted AP of only 1.97 and 11.34. A build-policy-matched KITTI-RetinaNet default-versus-aligned INT8 comparison improves aligned INT8 clean AP from 7.13 to 14.83 and mean corrupted AP from 7.43 to 12.07, but remains far from the reused FP8

treatment (49.27 clean AP and 36.48 mean corrupted AP), which was not rebuilt. These experiments demonstrate recipe-portability failure and bound the study’s generalization; they do not establish that INT8 is intrinsically less robust than FP8. S1 reports the rebuild contrast and secondary TIDE diagnostics.

5. Discussion

5.1. Absolute superiority and corruption sensitivity are different claims

The final holdout provides the clearest example of the paper’s central measurement distinction. FP8 begins 1.60 AP ahead of INT8 on original-clean images and remains 1.05 AP ahead after corruption, yet the clean-adjusted interaction is -0.55 AP. There is no contradiction: the absolute corrupted-input comparison favors FP8, while the interaction says that part of its pre-existing advantage is lost under corruption. The same pattern occurs at cell level in both the exploratory and selection-disjoint grids. Reporting only the corrupted gap would hide this change; reporting only the interaction would hide the fact that FP8 still retains higher corrupted AP on average.

The absolute-accuracy guardrail is equally important. Equation (4) shows that when both treatments approach zero corrupted AP, an initially positive clean gap mechanically yields a negative limiting interaction. A negative ΔE therefore cannot be interpreted as useful robustness without the corrupted AP values themselves. Conversely, relative retention can disagree in sign with the additive AP-point interaction because it normalizes by different clean denominators. These are not competing answers to one metric; they are different estimands and should be named explicitly.

5.2. What the paired protocol contributes

The practical contribution is the combination of measurement and execution constraints. Shared inputs ensure that INT8 and FP8 see the same condition-specific bytes; common image draws keep the four AP arms paired; the clean-control definition is explicit;

and absolute performance is reported before the interaction. The covariance audit shows why the shared resampling schedule is not cosmetic: analytically removing only the observed clean–corrupted covariance changes the estimated SD in most exploratory cells. The prediction-level worked example further connects the reported effect to an evaluator that reconstructs AP and uncertainty from retained predictions rather than from cached aggregate scores.

The controlled clean-input study clarifies another source of ambiguity. In the six final holdout blocks, original-source versus JPEG-95 clean substitution changes the interaction by only about +0.06 AP on average and its interval crosses zero. This reduces the weight of the historical codec mismatch as an explanation for the final result, but it does not prove codec invariance. The stronger methodological point is that control substitution can be isolated and estimated rather than left implicit.

5.3. Scope, generalization, and limitations

The primary empirical evidence is a YOLO11 TensorRT study. RT-DETR and RetinaNet do not extend the format comparison because their transferred INT8 recipes are not clean-competitive. The RetinaNet follow-up improves aligned INT8 but remains far from the frozen FP8 comparator, showing that matching the two INT8 builds’ recorded settings is insufficient for a balanced cross-family comparison. FP8 was not rebuilt under that matched policy. A stronger claim about architecture-independent INT8–FP8 robustness would require clean-competitive treatments for multiple detector families under matched build policies.

The corruption coverage is also bounded. Four synthetic corruption families and three severities do not represent all natural distribution shifts, and the corruption-realization branch contains only three frozen realization labels. The final post-selection rerun covers VOC and KITTI only. KITTI images are split by image identity rather than verified scene/drive identity, and converted VOC/KITTI annotations do not reproduce their official challenge rules. The seed studies cover YOLO11m on transfer datasets rather than the full grid, while fixed-class-universe resampling is investigated only for TT100K. These analyses therefore quantify conditionality of the recorded design rather than population variance over datasets, corruptions, or training procedures.

Reproducibility is similarly layered. Retained hashes, graph/build records, prediction bindings, compact ledgers, and a prediction-level example support stronger verification than aggregate tables alone, but they do not constitute independent regeneration of every historical detector run. Missing historical software metadata and unavailable payloads remain explicit limitations. Runtime measurements

cover TensorRT engines on one RTX 5090 and exclude end-to-end pipeline cost. Within these boundaries, the paired design applies beyond INT8/FP8: any two fixed detector treatments can be compared when a clean accuracy difference risks being conflated with sensitivity to input degradation.

6. Conclusion

A raw corrupted-input gap is not sufficient to characterize robustness when quantized detectors already differ on clean images. The proposed paired four-arm evaluation reports clean fidelity, remaining corrupted accuracy, and the corruption-associated change in the between-treatment gap under shared inputs and common image resampling. In selection-disjoint VOC/KITTI holdouts, FP8 retains a +1.05 AP mean corrupted advantage over INT8 while its clean-adjusted interaction is −0.55 AP, demonstrating that absolute superiority and corruption sensitivity can lead to different interpretations. The broader exploratory and sensitivity analyses show that control definition, metric scale, covariance, and treatment quality can further affect the conclusion. For quantized detector deployment studies, these quantities should be reported together rather than compressed into a single robustness ranking.

Data and code availability

The project repository provides analysis code, compact CSV/JSON ledgers, validation reports, and deterministic scripts for regenerating the retained summaries and figures. Release v2.2.0 is archived at <https://doi.org/10.5281/zenodo.22664869>; all archived versions are grouped under concept DOI <https://doi.org/10.5281/zenodo.22031663>. The repository is <https://github.com/iamdinhthuan/paired-floor-guarded-int8-fp8-detection>. Raw benchmark datasets remain subject to their original licenses and are not redistributed in the journal-source archive.

The v2.2.0 archive includes CVIU_Reviewer_Evidence.zip, separately from the compact journal-source ZIP. It contains the controlled clean-input experiment, recovered holdout artifact chains, updated AP/draw ledgers, frozen registries, validation records, implementation tests, and the KITTI/YOLO11m/fog-severity-1 prediction-level example. These follow-ups are not contained in the earlier v2.1.0 release. The evidence manifest binds the included manuscript sources and rendered PDFs; its README gives verification instructions and licensing details. Original software is MIT licensed, while KITTI-derived annotations retain CC BY-NC-SA 3.0 attribution, noncommercial and share-alike terms. Neither package provides full detector-inference regeneration. Source compilation,

summary recomputation, prediction-level evaluation, and full detector-inference regeneration therefore remain distinct reproducibility levels.

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Declaration of competing interest

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CRedit authorship contribution statement

Dinh Thuan Nguyen: Conceptualization, Methodology, Software, Investigation, Formal analysis, Writing—original draft. **Lam Phuong Nguyen:** Validation, Data curation, Writing—review and editing. **Vinh Huy Nguyen:** Methodology, Visualization, Writing—review and editing. **Sy Vu Quang:** Validation, Writing—review and editing. **Mohan Rajesh Elara:** Supervision, Writing—review and editing. **Anh Vu Le:** Supervision, Project administration, Writing—review and editing.

Declaration of generative AI and AI-assisted technologies in the writing process

During preparation of this work, the authors used OpenAI ChatGPT and OpenAI Codex for author-directed language revision, manuscript critique, coding assistance, experimental orchestration support, analysis implementation, validation, and consistency checking, and Anthropic Claude Code for manuscript critique and submission-file review. After using these tools, the authors reviewed, tested, and edited the resulting text or code as needed and take full responsibility for the content of the article. Numerical measurements reported in the paper were produced by the recorded experimental software, not by a generative-AI model.

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